

NASA: Human error caused loss of Mars orbiter

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WASHINGTON (AP) -- Failure to convert English measures to metric values caused the loss of the Mars Climate Orbiter, a spacecraft that smashed into the planet instead of reaching a safe orbit, a NASA investigation concluded Wednesday.

The Mars Climate Orbiter, a key craft in the space agency's exploration of the red planet, vanished after a rocket firing September 23 that was supposed to put the spacecraft on orbit around Mars.

An investigation board concluded that NASA engineers failed to convert English measures of rocket thrusts to newton, a metric system measuring rocket force. One English pound of force equals 4.45 newtons. A small difference between the two values caused the spacecraft to approach Mars at too low an altitude and the craft is thought to have smashed into the planet's atmosphere and was destroyed.

The spacecraft was to be a key part of the exploration of the planet. From its station about the red planet, the Mars Climate Orbiter was to relay signals from the Mars Polar Lander, which is scheduled to touch down on Mars next month.

"The root cause of the loss of the spacecraft was a failed translation of English units into metric units and a segment of ground-based, navigation-related mission software," said Arthus Stephenson, chairman of the investigation board.

Artist's conception of the Polar Lander on Mars

Stephenson said the board also found other facts that allowed the error to occur and to continue throughout a series of midcourse rocket firings.

The board found that the error went undetected in ground-based computers. Also, the mission navigation's team had an imperfect understanding of how the craft was pointed in space. Additionally, the mission navigation team, the report said, was overworked and not closely supervised by independent experts.

The Mars Climate Orbiter was launched December 11, 1998, and began its long journey toward the red planet. Along the way, engineers on the ground sent instructions to the craft to fire rockets to correct its path toward Mars. It was in these rocket firings that the error occurred.

On September 23 a final rocket firing was to put the craft into orbit, but the signals disappeared, leading engineers to believe that it traveled too close to Mars and was captured by the red planet's.

An investigation launched immediately quickly discovered that the mathematical conversion of English pounds to metric newtons had not been done.